

Liquidity in the U.S. Treasury Market

Nicholas Wong*

Stanford GSB

Haoxiang Zhu†

MIT Sloan School of Management

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Abstract

The U.S. Treasury market is often viewed as the most liquid financial market in the world. This paper examines whether that liquidity is uniform across trades. Using transaction-level TRACE data for on-the-run nominal coupon Treasuries, we organize evidence around the idea of bifurcated liquidity. Standardized electronic trading, especially around the \$1 million lot, appears extremely liquid. Outside that segment, transaction costs vary meaningfully across counterparties, venues, and trade sizes. The evidence is organized around cross-counterparty cost differences and the \$1 million standard-lot result. **Nick: Abstract carried over from the sketch; to be rewritten.**

* Stanford GSB. nicwjh@stanford.edu.

† MIT Sloan School of Management and NBER. zhuh@mit.edu.

1 Introduction

Nick: Key findings, placeholders for authoring.

- Claim: liquidity in the U.S. Treasury market is not uniform. It is excellent in standardized electronic trading, above all at the \$1 million lot, and materially worse outside it.
- TRACE (FINRA Treasury TRACE, September 2022 to September 2025, 214.5 million on-the-run trades in seven tenors) identifies counterparty type and ATS execution, so transaction costs can be measured at the trade level against a same-day inter-dealer ATS benchmark.
- Off ATS, customers pay a positive and significant premium over PTFs in every tenor, 0.15 to 1.02 cents per \$100 par at \$1 million (Table 4); on ATS the differences are far smaller (Table 5).
- The premium is a size phenomenon. At exactly \$1 million the customer cost level is within 0.05 cents of the inter-dealer benchmark in five of seven tenors, and customer minus PTF has no consistent sign (Tables 6 and 7 and fig. 1).
- The buy premium disappears at the standard lot: Buy is 0.16 to 1.34 cents off ATS across all sizes and zero or negative in six of seven tenors at exactly \$1 million.
- Venue use differs sharply by counterparty type (customers 0.1% ATS, PTFs 99%); on- and off-ATS costs are reported separately with no causal claim about venue.
- A carry-based interpretation of the customer buy-sell wedge is not supported on the full sample (??).
- Related literature paragraph: to be added once references are settled.
- Roadmap: Section 2 institutional setting; Section 3 data, filters and cost measures; Section 4 costs by counterparty; Section 5 the \$1 million standard lot; ?? carry; ?? robustness; Section 6 concludes.

2 Institutional Background

Nick: Core points, placeholders for authoring. Keep institutional detail short.

- The Treasury market has inter-dealer and dealer-to-customer segments; inter-dealer trading is heavily electronic.

- TRACE identifies counterparty type and whether a trade executed on an ATS. It does not identify the protocol within ATS (central limit order book against request for quote); inferences can be drawn but no FINRA documentation supports the distinction.
- Counterparty codes: P, principal trading firm; C, customer; A, non-member affiliate; T, ATS counterparty; D, inter-dealer; N, bank dealer; B, bank. N and B are reported from September 1, 2022, which sets the sample start.
- Inter-dealer trades executed on an ATS (D-D ATS=Y) form the benchmark for transaction-cost measurement.
- Seven on-the-run tenors: 2, 3, 5, 7, 10, 20 and 30 years. The 2- through 7-year have a new issue every month; the 10-, 20- and 30-year a new issue each quarter, reopened in the intervening months.

3 Data and Transaction-Cost Measurement

3.1 Sample

- FINRA TRACE Enhanced Historic Time and Sales, on-the-run nominal coupon Treasuries (no bills, TIPS, FRNs, strips or off-the-run securities), seven tenors, September 1, 2022 to September 30, 2025.
- The start date gives the richest cross-section of counterparty types, including B and N.
- Attrition from raw records to the analysis sample is in Table 2: 432.4 million raw records, 214.5 million trades after all filters.

3.2 Filters

- Standard TRACE filters: original disseminated reports only (Dissemination Flag = Y), which removes cancellations, corrections and reversals and leaves each inter-dealer trade once; decimal price format only.
- Clean filter, exclusions: Trade Modifier 4 in {B, S, W}; Trade Modifier 2 equal to H; when-issued trades.
- Price screen: 70 to 130.
- A 2 to 6% yield screen applies only to the benchmark yields used in the carry analysis (??), not to the trade sample.

3.3 Benchmark and transaction-cost definitions

- Benchmark universe: D-D ATS=Y trades in the same CUSIP and trading day.
- Realized spread (primary, M3): match to the next D-D ATS=Y trade. Effective spread (robustness, M2): match to the most recent one.
- $TC_j = 100 \cdot s_j \cdot (P_j - P_j^{\text{bench}})$ with $s_j = +1$ for a dealer sell (counterparty buy) and -1 for a dealer buy; prices per \$100 par, so TC_j is in cents per \$100 par. Match coverage exceeds 99.9% in every tenor (Table 1).
- Trade size $q_j = \text{Quantity}_j / 1,000,000$. Regressions use $\log_{10} q_j$, so a size coefficient is the change in cost per tenfold increase in size and an intercept is the predicted cost at \$1 million, where $\log_{10} q_j = 0$.
- Standard-lot indicator $1\{\text{Quantity}_j = 1,000,000\}$; not_1m_j is its complement.
- Buy indicator: one when the dealer sells to the counterparty.
- Buy-side-only counterparties: Bank (B), bank dealer (N) and ATS counterparty (T) appear only as buyers, because where both parties report only the sell-side report is disseminated. Their levels in Tables 4 and 7 are read at Buy = 0, a transaction type that does not occur for them; the predicted cost of a trade they do is the counterparty coefficient plus the Buy coefficient.

Table 1. Sample summary by tenor

	CUSIPs	Trading days	Trades (M)	Par (\$B)	Trades/day	Par/day (\$B)	M3 match %	M2 match %
2y	38	664	27.7	58,098	41,772	87.5	99.97	99.98
3y	38	636	25.2	44,061	39,663	69.3	99.97	99.98
5y	38	693	58.2	106,903	83,972	154.3	99.98	99.99
7y	38	722	19.5	29,049	26,998	40.2	99.96	99.97
10y	13	730	55.2	93,154	75,572	127.6	99.99	99.99
20y	13	700	7.5	10,087	10,782	14.4	99.92	99.94
30y	13	740	21.2	28,716	28,590	38.8	99.97	99.98
Total	190	773	214.5	370,067	277,515	478.7	99.98	99.99

On-the-run nominal coupon Treasuries, September 1, 2022 to September 30, 2025, after the clean filter and the 70 to 130 price screen, all venues and counterparties. CUSIPs: distinct on-the-run securities in the tenor over the sample; the two-, three-, five- and seven-year have a new issue every month, the ten-, twenty- and thirty-year a new issue each quarter, reopened in the intervening months. Trading days: days with at least one trade in the tenor. Trades in millions; par in \$ billions. Match coverage: share of trades with a matched inter-dealer ATS benchmark for the realized (M3) and effective (M2) spread. The Total row's per-day figures divide by the 773 trading days in the sample.

Table 2. Sample attrition

	Trades	Par (\$B)	% of raw
Raw TRACE records	432,400,462	938,656	100.0
After standard filters	266,857,312	575,761	61.7
After execution-date window	266,784,575	575,465	61.7
After clean filter	224,847,213	386,632	52.0
After price screen	224,844,163	386,620	52.0
After sample window (from 2022-09-01)	214,519,147	370,067	49.6

Cumulative: each row applies its filter to the row above. Raw TRACE records are every disseminated and non-disseminated record in the on-the-run files. Standard filters: dissemination flag Y, decimal price, deduplication. Execution-date window: records whose execution date lies outside the quarter of the file are removed. Clean filter: excludes trades with Trade Modifier 4 in {B, S, W}, Trade Modifier 2 equal to H, and when-issued trades. Price screen: 70 to 130. Sample window: September 1, 2022 onward, the first processed quarter having begun on July 1, 2022. The yield screen used for the carry analysis is not applied to the trade sample.

Table 3. Counterparty and venue composition

	Trades (K)	Par (\$B)	ATS % n	ATS % par	\$1M % n	\$1M % par	Cost
PTF	67,923.1	87,815	99.3	99.3	84.5	65.4	0.104
Customer	10,741.1	46,388	0.1	0.3	48.1	11.1	0.285
Affiliate	4,798.7	21,923	0.0	0.0	43.1	9.4	0.300
ATS counterparty	49,119.2	73,717	100.0	100.0	76.6	51.1	-0.032
Inter-dealer	74,853.6	124,105	63.4	56.5	72.2	43.5	N/A
Bank	45.9	364	65.1	15.3	47.4	6.0	0.283
Bank dealer	7,037.6	15,753	56.8	41.4	70.3	31.4	0.170
Total	214,519.1	370,067	78.3	64.2	75.2	43.6	0.119

Full sample, September 1, 2022 to September 30, 2025, after the clean filter and the 70 to 130 price screen, all venues. Trades in thousands. % n is the share of the counterparty's trade count; % par the share of its par volume. Cost is the par-weighted mean realized spread (M3) over matched trades, in cents per \$100 par: an unconditional mean across all sizes and directions, not comparable to the counterparty regression levels, which are read at \$1 million par and Buy = 0. Inter-dealer (D) trades define the benchmark and carry no spread. The Total cost is the par-weighted mean over the six counterparties with a spread.

Nick: Key findings, placeholders for authoring.

- \$1 million is the modal trade size for every counterparty: 48% of customer trades and 43% of affiliate trades are exactly \$1 million (Table 3).
- Customers execute about 0.1% of trades on ATS and PTFs about 99%, so venue and counterparty are nearly collinear for those two types.

- The Table 3 cost column is an unconditional par-weighted mean over all sizes and directions; it is not comparable to the Table 4 levels, which are read at \$1 million and Buy = 0.
- The monthly-auction tenors have fewer trading days than the quarterly ones (664 to 722 against 700 to 740 of 773). **Nick: Explanation to be confirmed: the on-the-run security changes each month and trades when-issued until settlement, which the clean filter excludes.**

4 Transaction Costs Differ by Counterparty

- Specification, off ATS: $TC_j = b_1 \log_{10} q_j + b_2 \text{Buy}_j + \sum_k \delta_k 1\{\text{type}_j = k\} + e_j$, no constant and no omitted category, so each δ_k is the predicted cost level of counterparty k at \$1 million par and Buy = 0, in cents per \$100 par. Algebraically equivalent to omitting PTF, but a thin PTF cell (PTF has almost no off-ATS trades before March 2025) no longer inflates the other standard errors.
- Counterparties: P, C, A, B, N off ATS; T added on ATS. D-D ATS=Y trades are excluded since they define the benchmark.
- Standard errors clustered by trading date. Stars on the counterparty rows test whether the level differs from zero, not from another counterparty.

Table 4. Off-ATS, realized spread (M3), full sample

	2y	3y	5y	7y	10y	20y	30y
log10(size)	-1.1632*** (0.0951)	-0.9590*** (0.1051)	-0.6480*** (0.1245)	-0.8428*** (0.1231)	-1.0859*** (0.0755)	-3.2758*** (0.3990)	-1.4182*** (0.3497)
Buy	0.6484*** (0.0380)	0.5377*** (0.0323)	0.1555*** (0.0562)	0.3725*** (0.0372)	0.4098*** (0.0334)	1.3377*** (0.1897)	0.3134*** (0.1163)
PTF	-0.2466*** (0.0161)	0.0144 (0.1450)	0.1385 (0.1475)	-0.1752*** (0.0181)	-0.1133** (0.0441)	-0.6603*** (0.0968)	-0.0963 (0.0650)
Customer	0.4650*** (0.0250)	0.2924*** (0.0272)	0.2869*** (0.0564)	0.1182*** (0.0187)	0.4116*** (0.0376)	0.3620*** (0.0799)	0.7861*** (0.1266)
Affiliate	-0.0557** (0.0254)	-0.0999*** (0.0204)	0.0872** (0.0375)	0.0326 (0.0317)	0.0947*** (0.0213)	-0.8490*** (0.1761)	0.1766*** (0.0673)
Bank	-1.9945*** (0.0914)	-1.5757*** (0.1164)	-0.3565*** (0.0859)	-1.0792*** (0.1459)	-0.9668*** (0.1493)	0.0860† (3.0854)	-0.3121 (0.4036)
Bank dealer	-0.4646*** (0.0376)	-0.3723*** (0.0259)	0.0318 (0.0612)	-0.2329*** (0.0316)	-0.1309*** (0.0272)	-1.3184*** (0.1978)	-0.0863 (0.1030)
N	2,294,492	1,719,710	5,209,653	1,518,648	5,511,383	790,504	2,032,221
Clusters	659	635	689	721	730	699	740
R-squared	0.0190	0.0150	0.0034	0.0078	0.0051	0.0288	0.0041

Coefficients are cost levels in cents per \$100 par, read at \$1 million par and at Buy = 0. No constant and no omitted category. The $\log_{10}(\text{size})$ row is the slope in cents per tenfold increase in par, with par in \$1 million units. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the cost level differs from zero, not whether it differs from another counterparty: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations, estimable but not interpretable. Bank, Bank dealer and ATS counterparty are 100% buy in the deduplicated data, so their levels are extrapolated on direction.

Table 5. On-ATS, realized spread (M3), full sample

	2y	3y	5y	7y	10y	20y	30y
log10(size)	-0.0401 (0.0349)	-0.0263 (0.0250)	-0.0025 (0.0130)	-0.0240 (0.0526)	-0.0698*** (0.0215)	-0.0420 (0.1248)	0.0468 (0.1519)
Buy	-0.0627** (0.0296)	-0.0114 (0.0112)	-0.0137 (0.0094)	0.0028 (0.0102)	-0.0029 (0.0067)	-0.1317*** (0.0160)	-0.0228 (0.0154)
PTF	0.1480** (0.0718)	0.2579*** (0.0964)	0.1133** (0.0545)	0.0696** (0.0295)	0.0835*** (0.0306)	0.0436*** (0.0103)	0.1122** (0.0516)
Customer	1.4193*** (0.1684)	1.1053*** (0.1310)	1.3916 (0.9963)	0.5703*** (0.1844)	1.2950 (0.9239)	-0.9952 (1.5052)	0.5307** (0.2430)
Affiliate	-2.3911**† (1.2195)	-0.0248† (0.3924)	1.7141† (2.5426)	0.0103† (0.0296)	1.5776***† (0.0074)	–	–
ATS counterparty	0.0379* (0.0224)	0.0042 (0.0131)	0.0021 (0.0071)	-0.0044 (0.0097)	-0.0081 (0.0068)	0.1015*** (0.0134)	0.0036 (0.0146)
Bank	0.0985** (0.0385)	0.0391 (0.0283)	0.1182*** (0.0183)	0.1198** (0.0600)	0.1598*** (0.0194)	0.7412*† (0.4401)	0.2964*** (0.0637)
Bank dealer	0.0805 (0.0543)	0.1212** (0.0509)	0.0796*** (0.0280)	0.1012*** (0.0252)	0.1340*** (0.0413)	0.2533*** (0.0218)	0.1857*** (0.0254)
N	16,222,330	15,065,166	32,802,126	11,159,818	30,910,326	3,624,465	10,771,294
Clusters	659	635	689	722	730	699	740
R-squared	0.0000	0.0001	0.0000	0.0000	0.0000	0.0002	0.0000

Coefficients are cost levels in cents per \$100 par, read at \$1 million par and at Buy = 0. No constant and no omitted category. The $\log_{10}(\text{size})$ row is the slope in cents per tenfold increase in par, with par in \$1 million units. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the cost level differs from zero, not whether it differs from another counterparty: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations, estimable but not interpretable. Bank, Bank dealer and ATS counterparty are 100% buy in the deduplicated data, so their levels are extrapolated on direction.

Nick: Key findings, placeholders for authoring (full sample, Table 4; Memos 21 and 22).

- Off ATS, customers pay significantly more than PTFs in every tenor: the customer level is 0.12 to 0.79 cents and customer minus PTF is 0.15 to 1.02 cents, largest at the twenty- and thirty-year.
- PTF levels are within 0.2 cents of the inter-dealer benchmark in five tenors; minus 0.25 at the two-year and minus 0.66 at the twenty-year.
- Smaller trades cost more: the $\log_{10}(\text{size})$ slope is minus 0.65 to minus 3.28 cents per tenfold increase in par, so a \$100,000 trade costs about 0.6 to 3.3 cents more than a \$1 million one.
- Customer buys (dealer sells) cost more than customer sells: Buy is 0.16 to 1.34 cents, larger at the long end.

- Affiliate levels are close to zero (minus 0.10 to 0.18 cents, minus 0.85 at the twenty-year), while affiliates pay slightly more than customers in Table 3 (0.300 against 0.285). Working hypothesis, to be tested before any interpretation is written: a size-mix effect, since 43% of affiliate trades are exactly \$1 million against 48% of customer trades.
- Bank's large negative levels (minus 2.0 at the two-year) are an out-of-sample projection: banks trade at a \$110,000 median with 2% of trades at \$1 million. At exactly \$1 million the two-year Bank level is minus 0.28 (Table 7).
- On ATS (Table 5) cross-counterparty differences are far smaller; customers execute 0.1% of their trades there, so the on-ATS customer level is imprecise. **Nick: Magnitudes to be read from Table 5.**
- The cross-counterparty ordering is stable across the three sub-periods (Memo 22); sub-period tables follow the first draft.

5 The \$1 Million Standard-Lot Result

- \$1 million is the standard electronic lot in the Treasury market. This section shows that the customer premium of Section 4 lives outside it.
- Figure 1: par-weighted realized spread by size bucket, customer and PTF, all venues, four core tenors (all seven computed).
- Table 6: customer off-ATS trades, $TC_j = a + \text{not_1m}_j (b_1 + b_2 \log_{10} q_j) + e_j$. a is the cost at exactly \$1 million; $b_1 + b_2 \log_{10} q_j$ is the premium for not being a standard lot. No separate $\log_{10} q$ term (collinear) and no Buy dummy.
- Table 7: off-ATS trades of exactly \$1 million, $TC_j = b \text{Buy}_j + \sum_k \delta_k 1\{\text{type}_j = k\} + e_j$, the Table 4 form without size. Does a customer premium survive at the standard lot, and does the buy premium?

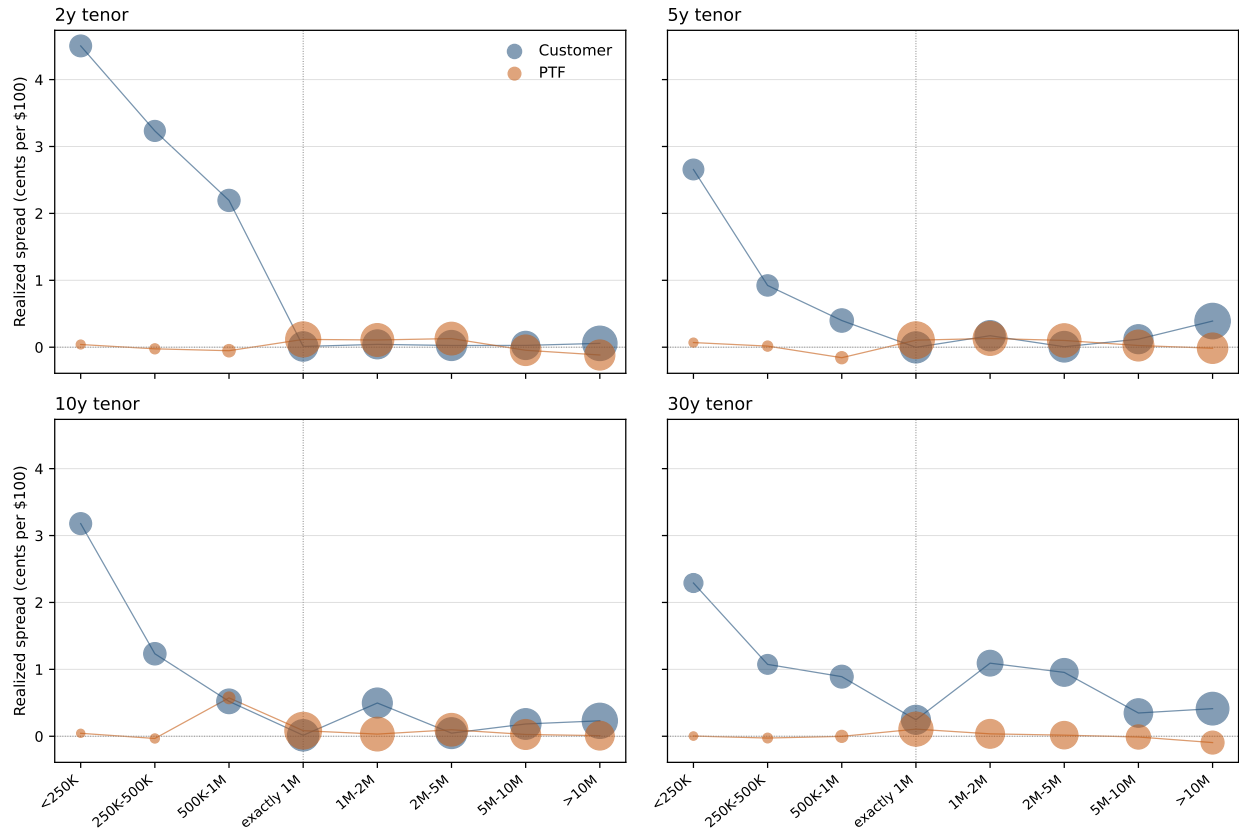


Figure 1. Transaction cost by trade size, customer versus PTF, four core tenors

Par-weighted mean realized spread (M3) within each size bucket, in cents per \$100 par, September 2022 to September 2025, all venues. Circle radius is proportional to $\log_{10}$ of par traded in the bucket. Customers execute about 0.1% of trades on ATS platforms and PTFs about 99%, so the customer series is effectively off-ATS and the PTF series effectively on-ATS: the figure compares venues as much as counterparties. The dotted vertical line marks the exactly-\$1 million bucket.

Table 6. Customer transaction costs: the \$1 million standard-lot interaction, realized spread (M3), full sample

	2y	3y	5y	7y	10y	20y	30y
a: cost at exactly \$1M	0.0109 (0.0161)	0.0072*** (0.0023)	-0.0025 (0.0049)	0.0002 (0.0022)	0.0120 (0.0127)	0.0484*** (0.0095)	0.2457*** (0.0633)
b1: not_1m	1.2943*** (0.0490)	1.1017*** (0.0817)	0.7282*** (0.0961)	0.8594*** (0.0924)	1.1438*** (0.0673)	1.7147*** (0.1319)	1.4393*** (0.2787)
b2: not_1m x log10(size)	-1.3937*** (0.1373)	-1.1686*** (0.1623)	-0.8566*** (0.1778)	-0.9091*** (0.1722)	-1.2909*** (0.1168)	-3.6013*** (0.5146)	-1.5699*** (0.5047)
N	1,532,131	1,125,320	2,521,407	1,127,567	2,941,608	560,987	917,904
N at \$1M	566,547	541,916	1,243,515	720,059	1,374,379	243,691	477,085
Clusters	659	634	689	720	730	697	740
R-squared	0.0215	0.0235	0.0062	0.0125	0.0071	0.0313	0.0052

Customer off-ATS trades, all sizes. Specification: $TC = a + \text{not_1m} (b_1 + b_2 \log_{10} q)$, with q par in \$1 million units, so $\log_{10} q = 0$ at exactly \$1 million. a is the cost of an exactly-\$1 million customer trade; $b_1 + b_2 \log_{10} q$ is the premium a trade of size q pays for not being a standard lot, so b_2 is the size slope among non-\$1 million trades in cents per tenfold increase in par. There is no separate $\log_{10} q$ term: it is identically zero at \$1 million and equal to the interaction elsewhere, so it is collinear. No Buy dummy, by design. Cents per \$100 par. Standard errors in parentheses, clustered by trading date.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7. Cross-counterparty costs at exactly \$1 million, off-ATS, realized spread (M3), full sample

	2y	3y	5y	7y	10y	20y	30y
Buy	-0.0530 (0.0389)	-0.0016 (0.0121)	-0.0568 (0.0359)	0.0114* (0.0067)	-0.0668*** (0.0252)	-0.2250*** (0.0408)	-0.1548** (0.0607)
PTF	0.0020 (0.0102)	0.0981 (0.0863)	0.2403 (0.1812)	-0.0079 (0.0053)	0.0842* (0.0493)	0.0667** (0.0332)	0.1422*** (0.0391)
Customer	0.0374*** (0.0135)	0.0080 (0.0058)	0.0252 (0.0185)	-0.0053* (0.0031)	0.0455** (0.0198)	0.1566*** (0.0211)	0.3200*** (0.0873)
Affiliate	0.0469** (0.0223)	0.0091 (0.0154)	0.0361* (0.0194)	0.0170 (0.0234)	0.0667*** (0.0219)	0.0541 (0.0926)	0.1987*** (0.0490)
Bank	-0.2754***† (0.0655)	-0.0086† (0.0517)	0.0666† (0.0514)	-0.0551† (0.2190)	0.1564***† (0.0465)	13.8956† (17.2364)	0.3934***† (0.1226)
Bank dealer	0.0858** (0.0389)	0.0417*** (0.0123)	0.1292*** (0.0337)	0.0867*** (0.0097)	0.1987*** (0.0250)	0.3860*** (0.0446)	0.3094*** (0.0571)
N	900,626	838,811	2,657,884	962,396	2,651,142	359,961	1,240,723
Clusters	657	632	689	718	730	698	740
R-squared	0.0000	0.0001	0.0001	0.0007	0.0001	0.0029	0.0000

Off-ATS trades of exactly \$1 million par. Specification: $TC = b \text{Buy} + \sum_k \delta_k 1\{\text{type} = k\}$. No constant and no omitted category, as in the all-sizes counterparty regression; no $\log_{10} q$, since every trade is the same size. Each δ_k is the cost level for counterparty k at $\text{Buy} = 0$, in cents per \$100 par. Bank and Bank dealer are 100% buy in the deduplicated data, so their levels are extrapolated on direction; the interpretable quantity for those rows is $\delta_k + b$. ATS counterparty has no off-ATS trades by construction. PTF off-ATS trading is concentrated after March 2025, so the PTF row is effectively a 2025 estimate. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the level differs from zero: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations; the Bank cells hold 9 to 372 trades and are not interpretable.

Nick: Key findings, placeholders for authoring (full sample, Memo 23).

- Figure 1: a trough at exactly \$1 million in all seven tenors. Customer cost falls from 2.3 to 4.5 cents below \$250,000 to 0.00 to 0.05 at \$1 million (0.25 at the thirty-year) and stays low above it. PTF stays within 0.3 cents of zero in every bucket but two. The thirty-year shows a second hump near 1 cent between \$1 million and \$5 million.
- Table 6: a is 0.00 to 0.01 cents in the two- through ten-year, 0.05 at the twenty-year and 0.25 at the thirty-year; b_1 is 0.73 to 1.71 cents, significant at 1% everywhere; b_2 is negative everywhere (minus 0.86 to minus 3.60 per tenfold increase in par), so the non-standard-lot premium is largest for the smallest trades.
- Table 7: at exactly \$1 million, Buy is zero or negative in six of seven tenors and significantly

negative at the ten-, twenty- and thirty-year, against 0.16 to 1.34 in Table 4. The seven-year is the only tenor with a positive Buy (0.01).

- Table 7: customer minus PTF is between minus 0.22 and plus 0.18 cents with mixed signs, so there is no consistent customer premium at the standard lot. Customer levels are 0.00 to 0.05 cents in five of seven tenors.
- Emerging framing (September 18, 2026): customers may pay more because they trade small, not because of who they are. The negative Buy at \$1 million is consistent with the dealer receiving cash on a sell and earning the overnight rate, which exceeds the bond yield when carry is negative (80% of tenor-days in the sample).
- Affiliate: not analysed separately in this section (decision September 18, 2026); it remains a row in Tables 3, 4 and 7.

6 Conclusion

Nick: Placeholders for authoring.

- Treasury liquidity is not uniformly high. The standardized electronic segment is highly liquid, and the \$1 million standard lot is its core.
- The customer premium is concentrated in non-standard sizes; at exactly \$1 million it is gone, and so is the buy premium.
- Customers may pay more because they trade small rather than because of counterparty type.
- Limitations: public TRACE lacks firm identifiers; venue is not randomly assigned, so on- and off-ATS panels are descriptive comparisons; the carry test is reduced form and not supported on the full sample.

Online appendix for

Liquidity in the U.S. Treasury Market

Nicholas Wong
Stanford GSB

Haoxiang Zhu
MIT Sloan School of Management

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[Latest Version](#) 

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Appendix A. Pooled panel

Table A.1. Pooled, realized spread (M3), full sample

	2y	3y	5y	7y	10y	20y	30y
log10(size)	-0.8076*** (0.0499)	-0.6817*** (0.0543)	-0.4265*** (0.0779)	-0.6819*** (0.0758)	-0.8675*** (0.0493)	-3.0507*** (0.3390)	-1.2802*** (0.3009)
Buy	0.0915*** (0.0298)	0.0827*** (0.0105)	0.0258* (0.0148)	0.0739*** (0.0109)	0.0939*** (0.0087)	0.3356*** (0.0584)	0.0665** (0.0307)
ATS	0.0724* (0.0371)	0.0591 (0.0522)	-0.0275 (0.0283)	0.0614** (0.0306)	0.0001 (0.0402)	0.0852** (0.0368)	-0.0117 (0.0275)
PTF	0.0760** (0.0384)	0.1931** (0.0824)	0.1520*** (0.0568)	-0.0014 (0.0151)	0.0827*** (0.0207)	-0.1953*** (0.0464)	0.1089*** (0.0420)
Customer	0.8774*** (0.0326)	0.5954*** (0.0336)	0.3646*** (0.0472)	0.2967*** (0.0238)	0.6149*** (0.0344)	1.0439*** (0.0808)	0.9333*** (0.1379)
Affiliate	0.1936*** (0.0207)	0.1111*** (0.0179)	0.1185*** (0.0186)	0.1698*** (0.0297)	0.2293*** (0.0172)	-0.2502** (0.1097)	0.3040*** (0.0305)
ATS counterparty	-0.0782** (0.0306)	-0.0871 (0.0571)	0.0392 (0.0347)	-0.0995*** (0.0366)	-0.0282 (0.0412)	-0.3980*** (0.0705)	-0.0410 (0.0327)
Bank	-0.8583*** (0.0524)	-0.6167*** (0.0695)	0.1178*** (0.0356)	-0.2562*** (0.0690)	0.0271 (0.0359)	0.7288† (1.4345)	0.2576*** (0.0871)
Bank dealer	0.0018 (0.0306)	0.0444* (0.0263)	0.1230*** (0.0265)	0.0322* (0.0172)	0.1366*** (0.0109)	-0.2408*** (0.0678)	0.1495*** (0.0214)
N	18,516,822	16,784,876	38,011,779	12,678,466	36,421,709	4,414,969	12,803,515
Clusters	659	635	689	722	730	699	740
R-squared	0.0012	0.0003	0.0003	0.0006	0.0012	0.0275	0.0016

Coefficients are cost levels in cents per \$100 par, read at \$1 million par and at Buy = 0. No constant and no omitted category. The $\log_{10}(\text{size})$ row is the slope in cents per tenfold increase in par, with par in \$1 million units. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the cost level differs from zero, not whether it differs from another counterparty: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations, estimable but not interpretable. Bank, Bank dealer and ATS counterparty are 100% buy in the deduplicated data, so their levels are extrapolated on direction.

Appendix B. Effective spread (M2)

Table B.1. Off-ATS, effective spread (M2), full sample

	2y	3y	5y	7y	10y	20y	30y
log10(size)	-1.1669*** (0.0832)	-0.9416*** (0.1072)	-0.6274*** (0.1273)	-0.8316*** (0.1241)	-1.0749*** (0.0766)	-3.3102*** (0.3651)	-1.4114*** (0.3183)
Buy	0.6802*** (0.0264)	0.5455*** (0.0324)	0.1566*** (0.0581)	0.3780*** (0.0371)	0.4464*** (0.0328)	1.3032*** (0.1712)	0.3609*** (0.1085)
PTF	-0.2497*** (0.0165)	0.0110 (0.1446)	0.1394 (0.1535)	-0.1611*** (0.0188)	-0.1276*** (0.0411)	-0.6255*** (0.0897)	-0.2691 (0.2062)
Customer	0.4581*** (0.0222)	0.2874*** (0.0277)	0.2798*** (0.0567)	0.1065*** (0.0191)	0.3863*** (0.0370)	0.3829*** (0.0718)	0.7461*** (0.1209)
Affiliate	-0.0687*** (0.0191)	-0.1097*** (0.0207)	0.0858** (0.0388)	0.0391 (0.0324)	0.0706*** (0.0199)	-0.8740*** (0.1623)	0.1567** (0.0624)
Bank	-2.0288*** (0.0798)	-1.5523*** (0.1188)	-0.3537*** (0.0877)	-1.0592*** (0.1454)	-0.9354*** (0.1440)	0.2693† (3.0649)	-0.2453 (0.3804)
Bank dealer	-0.4843*** (0.0278)	-0.3746*** (0.0258)	0.0339 (0.0629)	-0.2318*** (0.0316)	-0.1522*** (0.0282)	-1.2486*** (0.1795)	-0.1429 (0.0978)
N	2,295,025	1,720,636	5,210,561	1,519,268	5,512,173	790,959	2,033,172
Clusters	659	635	689	721	730	699	740
R-squared	0.0247	0.0147	0.0030	0.0076	0.0052	0.0293	0.0042

Coefficients are cost levels in cents per \$100 par, read at \$1 million par and at Buy = 0. No constant and no omitted category. The $\log_{10}(\text{size})$ row is the slope in cents per tenfold increase in par, with par in \$1 million units. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the cost level differs from zero, not whether it differs from another counterparty: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations, estimable but not interpretable. Bank, Bank dealer and ATS counterparty are 100% buy in the deduplicated data, so their levels are extrapolated on direction.

Table B.2. On-ATS, effective spread (M2), full sample

	2y	3y	5y	7y	10y	20y	30y
log10(size)	-0.0143 (0.0140)	-0.0264 (0.0250)	-0.0034 (0.0133)	-0.0146 (0.0523)	-0.0586*** (0.0206)	-0.0648 (0.1245)	0.0681 (0.1523)
Buy	-0.0265** (0.0133)	0.0063 (0.0118)	0.0194** (0.0095)	0.0384*** (0.0108)	0.0568*** (0.0063)	-0.0993*** (0.0223)	0.0401*** (0.0154)
PTF	0.1347** (0.0668)	0.2512*** (0.0962)	0.1028* (0.0545)	0.0567* (0.0295)	0.0630** (0.0303)	0.0224* (0.0120)	0.0895* (0.0514)
Customer	1.4297*** (0.1699)	1.0916*** (0.1313)	1.3754 (0.9962)	0.4769*** (0.1794)	1.3104 (0.9071)	-0.9412 (1.5081)	0.5563** (0.2421)
Affiliate	-2.3699**† (1.1953)	-0.0380† (0.3925)	1.7786† (2.5696)	-0.0305† (0.0290)	0.3327† (0.4172)	–	–
ATS counterparty	0.0101 (0.0126)	-0.0091 (0.0137)	-0.0215*** (0.0073)	-0.0324*** (0.0102)	-0.0509*** (0.0067)	0.0681*** (0.0204)	-0.0462*** (0.0149)
Bank	0.0257 (0.0173)	-0.0138 (0.0317)	0.0469** (0.0223)	-0.0313 (0.0716)	0.0436* (0.0241)	-0.1686† (0.4236)	-0.0166 (0.0932)
Bank dealer	0.0462 (0.0401)	0.1048** (0.0510)	0.0468* (0.0283)	0.0344 (0.0251)	0.0899** (0.0425)	0.2029*** (0.0267)	0.1319*** (0.0249)
N	16,223,549	15,066,639	32,803,662	11,160,566	30,911,451	3,624,667	10,772,314
Clusters	659	635	689	722	730	699	740
R-squared	0.0000	0.0001	0.0000	0.0000	0.0000	0.0001	0.0000

Coefficients are cost levels in cents per \$100 par, read at \$1 million par and at Buy = 0. No constant and no omitted category. The $\log_{10}(\text{size})$ row is the slope in cents per tenfold increase in par, with par in \$1 million units. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the cost level differs from zero, not whether it differs from another counterparty: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations, estimable but not interpretable. Bank, Bank dealer and ATS counterparty are 100% buy in the deduplicated data, so their levels are extrapolated on direction.

Table B.3. Pooled, effective spread (M2), full sample

	2y	3y	5y	7y	10y	20y	30y
log10(size)	-0.8004*** (0.0390)	-0.6698*** (0.0558)	-0.4120*** (0.0794)	-0.6690*** (0.0762)	-0.8552*** (0.0504)	-3.0796*** (0.3083)	-1.2711*** (0.2722)
Buy	0.1278*** (0.0128)	0.0987*** (0.0110)	0.0529*** (0.0153)	0.1045*** (0.0109)	0.1488*** (0.0082)	0.3491*** (0.0534)	0.1257*** (0.0280)
ATS	0.0669 (0.0429)	0.0541 (0.0522)	-0.0316 (0.0289)	0.0342 (0.0305)	0.0013 (0.0398)	0.0427 (0.0371)	0.0170 (0.0375)
PTF	0.0699* (0.0358)	0.1915** (0.0823)	0.1473** (0.0578)	0.0150 (0.0155)	0.0633*** (0.0202)	-0.1651*** (0.0463)	0.0590 (0.0387)
Customer	0.8697*** (0.0300)	0.5848*** (0.0341)	0.3431*** (0.0478)	0.2717*** (0.0235)	0.5794*** (0.0345)	1.0388*** (0.0737)	0.8874*** (0.1283)
Affiliate	0.1762*** (0.0115)	0.0971*** (0.0184)	0.1044*** (0.0193)	0.1620*** (0.0300)	0.1954*** (0.0165)	-0.3011*** (0.1032)	0.2778*** (0.0290)
ATS counterparty	-0.0980** (0.0455)	-0.0949* (0.0576)	0.0239 (0.0360)	-0.0953*** (0.0366)	-0.0675 (0.0415)	-0.3698*** (0.0677)	-0.1156*** (0.0417)
Bank	-0.8975*** (0.0476)	-0.6242*** (0.0701)	0.0607 (0.0386)	-0.3275*** (0.0745)	-0.0603 (0.0396)	0.3345† (1.4348)	-0.0066 (0.1070)
Bank dealer	-0.0240 (0.0212)	0.0337 (0.0267)	0.0992*** (0.0274)	0.0009 (0.0175)	0.0963*** (0.0103)	-0.2269*** (0.0633)	0.0755*** (0.0225)
N	18,518,574	16,787,275	38,014,223	12,679,834	36,423,624	4,415,626	12,805,486
Clusters	659	635	689	722	730	699	740
R-squared	0.0014	0.0003	0.0002	0.0006	0.0012	0.0278	0.0016

Coefficients are cost levels in cents per \$100 par, read at \$1 million par and at Buy = 0. No constant and no omitted category. The $\log_{10}(\text{size})$ row is the slope in cents per tenfold increase in par, with par in \$1 million units. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the cost level differs from zero, not whether it differs from another counterparty: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations, estimable but not interpretable. Bank, Bank dealer and ATS counterparty are 100% buy in the deduplicated data, so their levels are extrapolated on direction.

Table B.4. Customer transaction costs: the \$1 million standard-lot interaction, effective spread (M2), full sample

	2y	3y	5y	7y	10y	20y	30y
a: cost at exactly \$1M	0.0121** (0.0049)	0.0085*** (0.0023)	0.0022 (0.0061)	-0.0004 (0.0026)	0.0038 (0.0133)	-0.0073 (0.0105)	0.1947*** (0.0625)
b1: not_1m	1.3106*** (0.0438)	1.0981*** (0.0825)	0.7061*** (0.0981)	0.8344*** (0.0916)	1.1490*** (0.0706)	1.8072*** (0.1217)	1.5137*** (0.2577)
b2: not_1m x log10(size)	-1.4049*** (0.1236)	-1.1516*** (0.1649)	-0.8327*** (0.1822)	-0.9049*** (0.1740)	-1.2874*** (0.1181)	-3.6442*** (0.4771)	-1.5687*** (0.4672)
N	1,532,157	1,125,514	2,521,598	1,127,583	2,941,758	561,094	917,942
N at \$1M	566,450	541,862	1,243,507	719,987	1,374,330	243,672	477,077
Clusters	659	634	689	720	730	697	740
R-squared	0.0285	0.0231	0.0056	0.0122	0.0073	0.0324	0.0055

Customer off-ATS trades, all sizes. Specification: $TC = a + \text{not_1m} (b_1 + b_2 \log_{10} q)$, with q par in \$1 million units, so $\log_{10} q = 0$ at exactly \$1 million. a is the cost of an exactly-\$1 million customer trade; $b_1 + b_2 \log_{10} q$ is the premium a trade of size q pays for not being a standard lot, so b_2 is the size slope among non-\$1 million trades in cents per tenfold increase in par. There is no separate $\log_{10} q$ term: it is identically zero at \$1 million and equal to the interaction elsewhere, so it is collinear. No Buy dummy, by design. Cents per \$100 par. Standard errors in parentheses, clustered by trading date.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5. Cross-counterparty costs at exactly \$1 million, off-ATS, effective spread (M2), full sample

	2y	3y	5y	7y	10y	20y	30y
Buy	-0.0304 (0.0292)	-0.0001 (0.0120)	-0.0494 (0.0334)	0.0083 (0.0074)	-0.0273 (0.0194)	-0.2853*** (0.0405)	-0.1433** (0.0559)
PTF	0.0064 (0.0069)	0.0998 (0.0862)	0.2472 (0.1841)	0.0116* (0.0068)	0.0716 (0.0460)	0.1203*** (0.0344)	-0.0159 (0.1967)
Customer	0.0272** (0.0117)	0.0085 (0.0059)	0.0263 (0.0182)	-0.0044 (0.0037)	0.0175 (0.0170)	0.1299*** (0.0217)	0.2635*** (0.0837)
Affiliate	0.0322* (0.0176)	0.0016 (0.0156)	0.0274 (0.0186)	0.0058 (0.0242)	0.0355* (0.0196)	0.0291 (0.0931)	0.1803*** (0.0462)
Bank	-0.3217***† (0.0668)	-0.0403† (0.0461)	0.0348† (0.0545)	-0.2446***† (0.1170)	0.1511***† (0.0321)	13.9985† (17.1802)	0.4516***† (0.0950)
Bank dealer	0.0739** (0.0294)	0.0465*** (0.0123)	0.1276*** (0.0308)	0.0913*** (0.0102)	0.1590*** (0.0198)	0.5009*** (0.0441)	0.2883*** (0.0531)
N	900,560	838,781	2,657,851	962,392	2,651,045	359,970	1,240,791
Clusters	657	630	689	718	730	698	740
R-squared	0.0000	0.0001	0.0001	0.0006	0.0001	0.0041	0.0000

Off-ATS trades of exactly \$1 million par. Specification: $TC = b \text{Buy} + \sum_k \delta_k 1\{\text{type} = k\}$. No constant and no omitted category, as in the all-sizes counterparty regression; no $\log_{10} q$, since every trade is the same size. Each δ_k is the cost level for counterparty k at $\text{Buy} = 0$, in cents per \$100 par. Bank and Bank dealer are 100% buy in the deduplicated data, so their levels are extrapolated on direction; the interpretable quantity for those rows is $\delta_k + b$. ATS counterparty has no off-ATS trades by construction. PTF off-ATS trading is concentrated after March 2025, so the PTF row is effectively a 2025 estimate. Standard errors in parentheses, clustered by trading date. Stars on the counterparty rows test whether the level differs from zero: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A dagger marks a cell with fewer than 500 observations; the Bank cells hold 9 to 372 trades and are not interpretable.

Appendix C. Sub-period results

Nick: Sub-period tables follow the first draft. Periods: September 1, 2022 to December 31, 2023; calendar 2024; January 1 to September 30, 2025.